

Devesh Kumar

Software Engineer II | Azure, AI/LLM, IoT, Cloud-Native

 devesh1102@gmail.com
 +91 9308716025
 linkedin.com/in/devesh1102
 github.com/devesh1102
 devesh1102.github.io

SUMMARY

Software Engineer II at Microsoft with 4+ years building cloud-native platforms on Azure. Specializes in IoT data pipelines, knowledge graphs, and AI/LLM-powered copilots. Proven track record shipping production-grade microservices — from IaC and security hardening to Retrieval-Augmented Generation (RAG)-based agents and React front ends. First team across MCI to ship Azure OpenAI Assistant API in production; lifted Copilot query accuracy from 40% to 75%+ and architected a real-time vectorization pipeline avoiding projected 30–50% throughput slowdown.

SKILLS

• C# • Python • TypeScript • SQL / KQL • Bicep • C++ • Azure OpenAI • Azure Data Explorer • Cosmos DB • Event Hub • Redis • Kubernetes (AKS) • React • Fluent UI • Docker • Kafka • Playwright • Git • REST APIs • CI/CD • IaC (Bicep) • Managed Identity • RAG • Agile/Scrum • Threat Modeling

PROFESSIONAL EXPERIENCE

Software Engineer II | Microsoft India

Hyderabad, India

Jul 2021 – Present

Power Automate Process Mining

May 2025 – Present

- Implemented the OCPM (Object-Centric Process Mining) overview page from ground up, an edge-filter panel with infinite scroll, and five analytics widgets (bar, pie, line, KPI, activity card) in **React, Fluent UI, and TypeScript**; showcased at PPCC 2025.
- Made each analytics widget fully configurable — customers can remap the underlying metric each widget represents, unlocking **20–30 distinct configuration combinations** per widget without code changes.
- Redesigned the process details page to differentiate object-centric from case-centric processes; conducted a **Goal-to-Process POC** using Copilot Studio agent to convert business vocabulary rules into object-centric event log models with web-search for discovering missing entities.
- Authored comprehensive **Playwright end-to-end test suites** covering OCPM overview, all widgets, and filter panels — significantly cutting manual QA regression effort across the process mining platform.

AI Copilot and Manufacturing Agent

Nov 2023 – May 2025

- Integrated Copilot V3 with RAG retrieval, KQL-to-Cypher conversion, and typecast injection — **lifted query accuracy from 40% to 75%+**.
- First team across MCI** to ship Azure OpenAI Assistant API in production; spearheaded data-filtering, conversation threading, and file-search — demonstrated live at **HMI 2025 (Rolls Royce demo)** with 75%+ accuracy on customer datasets.
- Architected a real-time vectorization pipeline for entity and schema metadata via **dedicated Event Hub + Redis + Azure OpenAI** through Managed Identity, fully decoupled from main ingestion — **avoided projected 30–50% throughput slowdown**.
- Created **5 materialized ADX views** (Downtime, Actual/Scheduled Production & Consumption) as structured data slices for the assistant — eliminated complex traversal hallucinations.
- Ran entity decomposition POC achieving same Copilot accuracy with **50% fewer instructions and 62% fewer examples**; conducted Responsible AI and jailbreak testing, identifying real vulnerabilities submitted to the RAI Review process.
- Supported Parsec and Rolls Royce POCs end-to-end; **certified MCI Threat Model reviewer** — conducted security architecture review for the Industry AI team's DTT/DMF product.

Manufacturing Data Platform and Knowledge Graph

Jul 2021 – Oct 2023

- Led Python benchmarking of ADT, ADX, and Cosmos (parallel API calls, latency distributions); found **ADX 5–6x faster** than cold storage — drove ADT removal, **cutting node ingestion 39% (18→11 min)** and **relationship ingestion 54% (39→18 min)** for 330k nodes / 450k relationships.
- Designed multi-hop graph query APIs and time-series historization on **Azure Data Explorer**, enabling root-cause analysis across the manufacturing knowledge graph with configurable hops and path filters.
- Secured entire infrastructure via **virtual networks, private endpoints, and managed identity** across all Azure services (Cosmos DB, ADX, Redis, Storage) — eliminated all List Keys API usage.
- Developed Bicep IaC templates facilitating **one-click deployments**, slashing customer infrastructure provisioning time by 6 hours and elevating deployment success rate to 98%.
- Removed **1,152 code warnings** and enforced a **70% L0 test-coverage gate** in CI, improving overall codebase reliability.

Intern – Quant Research Team | Chicago Mercantile Exchange (CME) Group

Bangalore, India

May 2019 – Jul 2019

- Applied **Principal Component Analysis** to isolate interest rate curve movements and stress-tested margin risk models against extreme historical scenarios.
- Built an R tool to compute and report margin offsets at the currency level across all trading accounts.

EDUCATION

B.Tech / M.Tech (Dual Degree) — Electrical Engineering, Minor in Computer Science and Engineering

2016 – 2021

IIT Bombay | CGPA: 8.49 / 10

ACHIEVEMENTS

- Top 1% among 200,000+ aspirants in JEE Advanced 2016
- 99.1 percentile in JEE Mains 2016
- Qualified National Talent Search Exam (NTSE)